Batch Production Record: ER70 Steel Marine Propeller - Selective Laser Melting Process

Batch ID: BPR-SLM-2024-001
Product Description: Marine Propeller, ER70 Steel with 1 mm Machining Allowance
Production Date: October 1, 2024
Operator: John Doe
Supervisor: Jane Smith
Quality Control Inspector: Robert Brown
Equipment: SLM Machine ID: AM-07, Gas Atomizer ID: GA-12, CNC Mill ID: CNC-05

1.0 Batch Overview

This document records the complete production cycle for one (1) ER70 steel marine propeller manufactured using Selective Laser Melting (SLM) technology. The process involved three primary phases: metal powder production via gas atomization, additive manufacturing build, and finish machining. All materials, energy consumption, and outputs are logged per standard operating procedures.

Key Specifications: - Target Propeller Mass: 0.204 kg - Build Material: ER70 Steel Powder - Machining Allowance: 1 mm - Support Structures: Integrated into powder usage

Reference Data (Historical): - Previous batch (BPR-SLM-2024-000): Powder usage ~0.360 kg, SLM electricity ~12.1 kWh - Industry benchmark for similar SLM propellers: Machining waste ~0.110-0.130 kg

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2.0 Material Preparation Phase - Gas Atomization

Time Log: 08:00 - 10:15
Operator: John Doe
Equipment: Gas Atomizer GA-12
Material Lot: ER70 Billet Lot #B-8842

2.1 Process Notes

The gas atomization process commenced at 08:00 with pre-heating of the atomization chamber. ER70 steel billet was loaded into the melting crucible. Atomization parameters were set to standard operating conditions for marine-grade steel powder production.

08:30 - Melting initiated, billet consumption recorded.
09:15 - Atomization process active, argon shield gas flowing.
10:15 - Process completed, powder collection system engaged.

2.2 Resource Consumption

Raw material and utility consumption during gas atomization:

Resource Type	Quantity	Unit	Notes
Steel Billet	0.414	kg	ER70 grade, Lot #B-8842
Electricity	0.828	kWh	Atomization process power
Argon Gas	2.58	kg	High-purity, Cylinder #ARG-774
Cooling Water	0.116	kg	Closed-loop system makeup

Quality Note: Produced powder met sieve analysis specifications for SLM processing (15-45 μm).

3.0 SLM Build Phase

Time Log: 10:30 - 17:03
Operator: John Doe
Equipment: SLM Machine AM-07
Powder Lot: ER70 Powder Lot #P-9921 (from above atomization) **Build File:** Marine_Propeller_ER70_v3.2

3.1 Process Chronology

10:30 - Build plate installed and leveled. Powder recoater system calibrated.
10:45 - Build chamber purging initiated with argon.
11:00 - SLM build process started. Laser parameters set per ER70 material profile.
13:30 - Mid-build inspection - no anomalies observed.
17:03 - Build completion, chamber cooling initiated.

Process Parameters: **-** Total Build Time: 6.55 hours **-** Layer Thickness: 30 μm **-** Laser Power: 200W (average) **-** Scan Speed: 800 mm/s **-** Build Chamber Temperature: 80°C

3.2 Resource Consumption During Build

Resource Type	Quantity	Unit	Notes
ER70 Steel Powder	0.352	kg	Includes support structures
Electricity	11.49	kWh	SLM machine consumption
Compressed Air	3.56	m³	Filtered plant air, filtration system
Argon Gas	1.2	m³	Build chamber atmosphere

Equipment Performance: Machine AM-07 operated within specified parameters throughout build cycle. Average power draw during active build: 1.75 kW.

4.0 Post-Processing Phase - Finish Machining

Time Log: 17:30 - 21:38
Operator: John Doe
Equipment: CNC Mill CNC-05
Fixture: Custom propeller fixture #F-45

4.1 Machining Process

17:30 - Propeller transferred to CNC mill, fixture installation completed.
17:45 - Tool calibration and workpiece zeroing performed.
18:00 - Roughing operations commenced.
20:15 - Finishing passes initiated.
21:38 - Machining completed, part removed from fixture.

Machining Parameters: - Total Machining Time: 4.133 hours - Primary Operation: 1 mm allowance removal from all surfaces - Tooling: Carbide end mills (#T-334, #T-335) - Coolant System: High-pressure through-spindle

4.2 Resource Consumption During Machining

Resource Type	Quantity	Unit	Notes
Electricity	2.84	kWh	CNC mill and auxiliary systems
Cutting Fluid	1.94	kg	Synthetic water-miscible, Lot #CF-882

Machine Performance: Average power consumption during active machining: 0.687 kW. No tool changes required.

5.0 Output and Waste Summary

5.1 Production Output

21:45 - Final inspection of completed propeller.

Output Type	Quantity	Unit	Specifications
Marine Propeller	1	units	ER70 Steel, 0.204 kg
Final Product Mass	0.204	kg	Within tolerance ±0.005 kg

5.2 Waste Generation

Waste Type	Quantity	Unit	Disposition
Machined Chips	0.116	kg	ER70 steel, sent to recycling bin #R-07

Material Yield Calculation: - Total powder used: 0.352 kg - Final product: 0.204 kg - Machining waste: 0.116 kg - Unaccounted (support structures, overspray): 0.032 kg

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6.0 Quality Control Verification

Time: 22:00
Inspector: Robert Brown

6.1 Dimensional Inspection

All critical dimensions verified against drawing #MP-ER70-004: - Blade profile: Within ±0.1 mm tolerance - Hub dimensions: Within ±0.05 mm tolerance - Surface finish: Ra 6.3 μm (as-machined)

6.2 Visual Inspection

No visible defects, cracks, or surface imperfections observed. Support removal points properly dressed.

QC Status: APPROVED
Disposition: Ready for coating and final assembly

7.0 Batch Closure

Time: 22:15
Operator: John Doe
Supervisor: Jane Smith

All production data recorded accurately. Equipment shut down according to standard procedures. Production area cleaned and prepared for next shift.

Batch Status: COMPLETED SUCCESSFULLY
Next Steps: Transfer to coating department per work order #WO-8847

Signatures:

Operator: ___________________________
John Doe

Quality Control: ___________________________
Robert Brown

Production Supervisor: _________________________
Jane Smith

Date: October 1, 2024

End of Batch Production Record
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